Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/21/2025
PI Name:	Goukassian, David A M.D., Ph.D.		
Project Title:	Space Relevant Radiation-Induced Cardiovascular Disease Risk Thresholds: Effect of Sex on the Outcome		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI moved to Icahn School of Medicine at Mount Sinai from Temple University in October 2018.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	04/10/2019	**End Date:**	12/31/2026
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Continuation of "Space Relevant Radiation-Induced Cardiovascular Disease Risk Thresholds: Effect of Sex on the Outcome," grant 80NSSC18K0921, due to PI move to Icahn School of Medicine at Mount Sinai from Temple University. (Ed., 2/10/2020) NOTE: End date changed to 12/31/2026 per S. Mack-Phillips (Ed., 2/20/25) NOTE: End date changed to 02/15/2024 per NSSC information (Ed., 5/26/23)		
Key Personnel Changes/Previous PI:	PI notes additional collaborators who are assisting with the project: Lahouaria Hadri, Kenneth Walsh, and Venkata Naga SrikanthGarikpati (Ed., 5/26/23).		
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1079		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Ed. note 2/10/2020: Continuation of "Space Relevant Radiation-Induced Cardiovascular Disease Risk Thresholds: Effect of Sex on the Outcome," grant 80NSSC18K0921 with the same Principal Investigator Dr. David Goukassian, due to PI move to Icahn School of Medicine at Mount Sinai from Temple University. During the future Moon, near Earth asteroids, and Mars missions, astronauts will be exposed to higher total doses of space irradiation (IR) (~0.4-0.5 Gy) from galactic cosmic rays (GCR). Most of what we know about harmful effects of IR on cardiovascular (CV) system is from epidemiological studies of long-term survivors of cancer radiotherapy (RT). A recent study of 2,168 women who underwent RT for breast cancer has shown that the rates of major coronary events increased linearly with the mean dose to the heart by 7.4% per Gy, with no apparent lower or upper threshold. In this study, it was determined that average of the mean doses to the whole heart was 4.9 Gy with the range of 0.03 - 27.72 Gy. Furthermore, metabolomics studies, in patients undergoing hematopoietic stem cell (HSC) transplantation as part of cancer treatment (1.25 Gy total-body irradiated), identified seven urine-based biomarkers with distinct differences between pre- and post-exposure samples. The levels of these markers were found to be sex-dependent suggesting that separate biomarker signatures may exist for males and females. Hypotheses: Our central hypothesis is that low-dose proton and HZE (high energy) particle IR-induced biological responses are long-lasting, IR type- and dose-dependent and may augment excess relative risk (ERR) estimates for the development of cardiovascular diseases (CVDs) during and after long-duration space missions. In addition, we hypothesize that sex differences could further modify radio-biologically effective (RBE) IR thresholds for CV risk estimates. Gene expression and epigenetic modifications in protein and microRNA (miRNA) in exosomes from the blood (e.g., plasma/serum) may be altered before the onset of the cardiac symptoms, which could be used as potential biomarkers to predict the CVD risks. We will test our hypotheses with the following specific aims: AIM 1. Determine the longitudinal effect of IR type, dose, and sex on cardiovascular physiology in wild type mice and ApoE null mice after full-body 5-ion simplified mixed field and gamma radiation. AIM 2. Determine space-type IR mediated modulations in exosomal cargo in the blood, and determine whether these changes are associated with alterations in the heart function, structure, and vasculature before the manifestation of clinical symptoms. AIM 3. Utilize known and newly identified biomarkers in the blood to develop human-relevant point-of-care tests (POCT) for predicting and monitoring possible CV alterations before and during space flights. We anticipate that the results of our proposed work may be beneficial for human space exploration and could (1) Determine single whole body mixed field dose-response, radio-biologically effective IR thresholds in the heart and cardiac vasculature, and whether sex differences could modify radio-biologically effective IR thresholds for CV risk estimates; 2) Determine whether space radiation leads to modifications in the circulating exosomal cargo contents and whether IR-induced exosomal cargo modulations are reflective of subclinical changes in the cells and organs of origin; 3) Ascertain if modulations of exosomal cargo may be representative of chronic oxidative stress and inflammation and could serve as early biomarkers of IR-induced CVD initiation and progression; 4) Integrate physiological CV endpoint data sets with gene expression and epigenetic data to identify biomarkers in bio-fluids that could be used for prediction of asymptomatic CVD in the setting of space IR, which will include known early and intermediate biomarkers of cardiac damage, inflammation, and oxidative stress, as well as currently unknown novel radiation-associated cardiac biomarkers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We anticipate that the results of our work could be beneficial for human space exploration as well as for the Earth-based applications on several levels -- (1) determine whether low dose space-type and terrestrial IR may present an increased risk for CVD development during and after prolonged space missions, as well as after conventional and particle cancer radiotherapy; (2) determine the underlying molecular signaling of CV alterations; (3) identify biomarkers in the blood that could be used for prediction of asymptomatic CV disease, which will include known early and intermediate biomarkers of cardiac damage, as well as currently unknown novel cardiac biomarkers; (4) the identification of sub-clinical CV disease biomarkers that could be used for monitoring the effectiveness of mitigating factors for prevention and treatment of IR-induced CVDs in space and in Earth-bound civilian population, in general.
	Ed. note 2/10/2020: Continuation of "Space Relevant Radiation-Induced Cardiovascular Disease Risk Thresholds: Effect of Sex on the Outcome," grant 80NSSC18K0921 with the same Principal Investigator Dr. David Goukassian, due to PI move to Icahn School of Medicine at Mount Sinai from Temple University. In one of our studies that was published in the Cardiovascular Diabetology journal we investigated the long-term (i.e., murine lifetime) sex-specific gene expression patterns in LV tissue using a transcriptomic approach in male versus female WD-fed C57BL/6J mice (continuously for 125 days) at a relatively young age, from 300 through 425 days old. Long-term consumption of the Western Diet (WD) is a risk factor for the development of cardiovascular disease (CVD); however, there is a paucity of studies on the long-term effects of WD on the pathophysiology of CVD and sex-specific responses. Our study aimed to investigate the sex-specific pathophysiological changes in left ventricular (LV) function using transthoracic echocardiography (ECHO) and heart tissue transcriptomics in WD-fed C57BL/6J mice for 125 days, starting at the age of 300 through 425 days. In female mice, consumption of the WD diet showed long-term effects on LV structure, as well as possible development of HFpEF-like phenotype with compensatory cardiac structural changes later in life. In male mice, ECHO revealed the development of an HFrEF-like phenotype later in life without detectable structural alterations. This sex-associated dichotomy in LV structure and function was evident in the transcriptomic profile. Specifically, at 530-day WD-fed male mice exhibited differentially expressed genes (DEGs), which were overrepresented in pathways associated with endocrine function, cellular processes, signal transduction, and cardiomyopathies. At 750 days, WD-fed male mice exhibited dysregulation of several genes involved in various lipid, glucagon, and glutathione metabolic pathways. At 530 days, WD-fed female mice exhibited the most distinctive set of DEGs with an abundance of genes related to circadian rhythms. At 640 days, in WD-fed female mice altered DEGs were associated with cardiac energy metabolism and remodeling. Our study demonstrated distinct sex-specific and age-associated differences in cardiac structure, function, and transcriptome signature between WD-fed male and female mice.

Task Progress:	In another study that was published in the Scientific Reports journal, we aimed to evaluate the changes in transcriptome landscape in the LV of male and female C57BL6/J mice at 440/440 and 660/550 days after single-dose full-body exposure with γ- and simGCRsim-IR. Our goal was to determine the potential sex-dependent molecular changes induced by different IR-types and aging, providing a clearer understanding of the underlying molecular mechanisms and the long-lasting effects of these IR in the heart tissue. We reported previously heart failure with preserved ejection fraction cardiac phenotype 660 days following exposure to a single dose of a simplified galactic cosmic ray simulation (simGCRsim) only in males with functional and structural impairment in left ventricular (LV) function. This sex-based dichotomy prompted us to investigate sex-specific changes in the LV transcriptome in three-month-old male and female mice exposed to 137Cs-gamma- or simGCRsim-irradiation (IR). Non-IR male and female (10 each) mice served as controls. LVs were collected at 440/660- and 440/550 days post-IR, male and female, respectively. RNA sequencing, differential gene expression, and functional annotation were performed on tissues from 5 mice/group. Sex and post-IR time points had the greatest influence on gene expression surpassing the IR-type effects. SimGCRsim-IR showed more persistent transcriptome changes than γ-IR. Thus, the long-term persistence of IR effects on LV tissues could pose significant health risks requiring attention, in terms of early development of mitigating factors. Based on these results, it can be concluded that the effect of IR is long-lasting both in males and females in all age groups, with more prominent differences observed in males at 440 days post-IR. Deep-space exploration-type missions to the Moon (NASA's Artemis and Gateway programs), Mars, and beyond will introduce prolonged exposure to health hazards, including but not limited to isolation and confinement, sleep disruption, exposure to different types, and quantities of particle radiation (i.e., high energy and atomic number – HZE). The known effects of space travel on the cardiovascular system include dysrhythmias, altered vascular compliance, dysautonomia, and induction of a pro-inflammatory state. It is not known how these concerns, and other potentially unknown CVD risks, will manifest during and/or after exploration-type missions. We provide a comprehensive summary of the most prominent risk factors on cardiac physiology and their known effects in a review manuscript that included - 1) distance from Earth, 2) altered gravity, 3) space radiation, 4) isolation and confinement, 5) sleep deprivation/fragmentation, and 6) food/nutrition. Submitted for publication to the Journal of American College of Cardiology Advances for publication consideration. To date, most of the studies on the effect of terrestrial and space-type IR on cardiovascular alterations have been reported for the LV; whereas, alterations in right ventricular (RV) structure, function, and underlying molecular mechanisms have not been studied well. In the next study we aimed to assess the effect of simplified galactic cosmic ray simulated (simGCRsim) and gamma (γ) ionizing radiation (IR) on transcriptome changes in right ventricular (RV) tissue 16 to 22 months after a single low dose (0.5 Gy, 500 MeV/nucleon) full body exposure in C57BL/6J male and female mice. Our results reveal no differentially expressed genes (DEGs) in female mice and only 2 upregulated genes in the male mice exposed to γ-IR after 16 and 22 months post-IR. In contrast, exposure to simGCRsim-IR resulted in 4 DEGs in females and 371 DEGs in males suggesting longer lasting and sex-biased DEGs after simGCRsim-IR. Overrepresentation analysis of DEGs in male simGCRsim-IR animals revealed significant enrichment in pathways related to muscle contraction, hypertrophic cardiomyopathy, oxytocin release, and the regulation of actin cytoskeleton. Meanwhile, in the same study group, the pathways associated with neurodegenerative diseases such as Alzheimer's, Huntington's, and Parkinson's diseases, were also enriched. Our results suggested that 16-22 months after exposure the RV transcriptome exhibits distinct responses based on both the type of irradiation and sex. These studies are in the final stages of preparation for submission.

Bibliography Type:	Description: (Last Updated: 04/23/2026)
Abstracts for Journals and Proceedings	Roksana Zakharyan, Siras Hakobyan, Suren Davitavyan, Gisane Khachatryan, Tamara Sirunyan, Ani Stepanyan, Arsen Arakelyan, David A. Goukassian. "Lifetime effects of space radiation on left ventricle transcriptome in male and female mice." 48th European Radiation Research Society Meeting, Porto, Portugal, September 10-13, 2024. Abstracts. 48th European Radiation Research Society Meeting, Porto, Portugal, September 10-13, 2024. , Sep-2024
Abstracts for Journals and Proceedings	Siras Hakobyan, Ani Stepanyan, Roksana Zakharyan, Suren Davidavyan, Gisane Khachatryan, Tamara Sirunyan, Arsen Arakelyan, David A. Goukassian. "Right ventricle long-term transcriptome perturbations in response to whole-body gamma and simGCRsim irradiation." 48th European Radiation Research Society Meeting, Porto, Portugal, September 10-13, 2024. Abstracts. 48th European Radiation Research Society Meeting, Porto, Portugal, September 10-13, 2024. , Sep-2024
Abstracts for Journals and Proceedings	Arakelyan A, Davitavyan S, Brojakowska A, Stepanyan A, Zakharyan R, Hakobyan S, Sirunyan T, Khachatryan G, Khlgatian MK, Bisserier M, Zhang S, Hadri L, Garikipati VNS, Goukassian DA. "Association of radiation-induced somatic events in blood cells with tumorigenesis and cardiac dysfunction." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Stepanyan A, Brojakowska A, Zakharyan R, Hakobyan S, Davitavyan S, Sirunyan T, Khachatryan G, Khlgatian MK, Bisserier M, Zhang S, Sahoo S, Hadri L, Garikipati VNS, Arakelyan A, Goukassian DA. "Evaluating sex-specific responses to western diet across the lifespan: Effects on cardiac transcriptomic signatures." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Davitavyan S, Brojakowska A, Stepanyan A, Zakharyan R, Hakobyan S, Sirunyan T, Khachatryan G, Khlgatian MK, Bisserier M, Zhang S, Hadri L, Garikipati VNS, Arakelyan A, Goukassian DA. "Examination of radiation-induced somatic alterations in the x- and y-chromosomes of blood cells." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025

Articles in Peer-reviewed Journals	Rai AK, Sanghvi S, Muthukumaran NS, Chandrasekera D, Kadam A, Kishore J, Kyriazis ID, Tomar D, Ponnalagu D, Shettigar V, Khan M, Singh H, Goukassian D, Katare R, Garikipati VNS. "Role of mitochondrial ribosomal protein L7/L12 (MRPL12) in diabetic ischemic heart disease." Free Radic Biol Med. 2024 Sep;222:531-538. https://doi.org/10.1016/j.freeradbiomed.2024.07.003 ; PubMed PMID: 38977138 , Sep-2024
Articles in Peer-reviewed Journals	Tahimic CGT, Steczina S, Sebastian A, Hum NR, Abegaz M, Terada M, Cimini M, Goukassian DA, Schreurs AS, Hoban-Higgins TM, Fuller CA, Loots GG, Globus RK, Shirazi-Fard Y. "Simulated microgravity alters gene regulation linked to immunity and cardiovascular disease." Genes (Basel). 2024 Jul 24;15(8):975. https://doi.org/10.3390/genes15080975 ; PubMed PMID: 39202335; PubMed Central PMCID: PMC11353732 , Jul-2024
Articles in Peer-reviewed Journals	Rai AK, Muthukumaran NS, Nisini N, Lee T, Kyriazis ID, de Lucia C, Piedepalumbo M, Roy R, Uchida S, Drosatos K, Bisserier M, Katare R, Goukassian D, Kishore R, Garikipati VNS. "Transcriptome wide changes in long noncoding RNAs in diabetic ischemic heart disease." Cardiovasc Diabetol. 2024 Oct 17;23(1):365. https://doi.org/10.1186/s12933-024-02441-6 ; PubMed PMID: 39420368; PubMed Central PMCID: PMC11488282 , Oct-2024
Articles in Peer-reviewed Journals	Stepanyan A, Brojakowska A, Zakharyan R, Hakobyan S, Davitavyan S, Sirunyan T, Khachatryan G, Khlgatian MK, Bisserier M, Zhang S, Sahoo S, Hadri L, Rai A, Garikipati VNS, Arakelyan A, Goukassian DA. "Evaluating sex-specific responses to western diet across the lifespan: impact on cardiac function and transcriptomic signatures in C57BL/6J mice at 530 and 640/750 days of age." Cardiovasc Diabetol. 2024 Dec 28;23(1):454. https://doi.org/10.1186/s12933-024-02565-9 ; PubMed PMID: 39732652; PubMed Central PMCID: PMC11682651 PMID: 39732652 , Dec-2024